

Preferences for Job Amenities and Adverse Selection: Evidence from Survey Enumerators

Bailey Palmer¹

¹Department of Economics, University of California, Berkeley

Job Market Paper in Progress: Please do not cite or distribute

July 22, 2025

Abstract

Can the provision of certain job amenities induce adverse selection on workers? Private information held by workers on their own productivity alongside heterogeneous preferences for job amenities can generate a classic selection problem depending on the correlation between unobserved productivity and willingness to pay for amenities. Working with a Jordanian research firm that hires hundreds of female survey enumerators a year, I combine administrative firm records on worker productivity with a survey which elicited individual level willingness to pay for several context-relevant job amenities such as childcare and transportation to shed new light on this question. For all amenities studied except childcare, I find that the least productive workers have the highest willingness to pay. This negative correlation is particularly large for transportation, and cannot be fully explained by income, location, commuting costs, or any other observable. To assess the empirical magnitude of this adverse selection, I write down a novel monopsony model that allows for job amenities to increase labor supply and to differentially select more/less productive workers. The model shows that mandating firm-provided transportation while allowing wages to adjust downward freely would increase employment, due to workers' large WTP for the amenity, but would lower average worker productivity via negative selection. The net effect is a reduction in firm profit, suggesting that adverse selection may be one reason why job amenities are not provided by the labor market even when worker WTP is high.

1 Introduction - not ready yet

The labor market is a classic setting for economic models of asymmetric information like screening and adverse selection. Workers have heterogeneous preferences over jobs and hold private information on their own productivity. In particular, workers clearly have heterogeneous preferences over job amenities such as work from home, flexibility, etc. To the extent that these preferences for job amenities may be correlated with unobserved productivity, offering contracts with more of the compensation in terms of amenities could theoretically select more or less productive

workers depending on the direction of the correlation. For example, offers with more of the compensation in amenities could disproportionately attract low-productivity type workers if low-type workers are more willing to trade wages for amenities than high-type workers. Despite the canonical nature of this question, the potential for job amenities to induce adverse selection has not received much attention, largely due to data constraints.

This paper combines administrative records of workers’ productivity at a Jordanian firm with survey data on those workers’ willingness to pay for real-life job amenities including transportation, childcare, and preferred pay schemes. I show that the least productive workers have the highest willingness to pay (WTP) for job amenities. The negative correlation is statistically significant and generates enough adverse selection to make offering these job amenities to entry-level workers unprofitable. Recent work has theorized on the possibility of job amenities inducing adverse selection (Nekoei, 2024) or used natural experiments to show that certain job amenities do appear to attract less productive workers (Emanuel & Harrington, 2024), but this paper is the first to my knowledge to measure the joint distribution of productivity and preferences for job amenities and shed light on the adverse selection problem that arises.

I leveraged my professional relationship with the largest research firm in Jordan to access detailed productivity records of survey enumerators, who are entirely women due to cultural norms. Using records of the speed and quality of data collected by individual enumerators across several years of projects, I generate an inverse covariance weighted index of productivity following Anderson (2008). I then pair this productivity index with survey-based willingness to pay data gathered from an incentivized multiple price list over 4 real-life amenities which were indicated as desirable in qualitative interviews: childcare, transportation, fixed pay (vs productivity pay), and monthly pay (vs lump sum at the end of data collection). I find the largest negative correlation for transportation: 1 SD increase in the productivity index is associated with 13 percent lower WTP for transportation. Statistically significant negative correlations are also present for daily pay and monthly paychecks. There is a negative point estimate for childcare, but it does not rise to statistical significance, potentially due to the smaller sample size (WTP for childcare was only elicited for mothers of young children).

Several robustness checks confirm that the negative correlation detected is consistent with adverse selection. Regressions using principal components of productivity instead of the Anderson index show qualitatively similar results. Additionally, the results remain similar when controlling for firm tenure, which rules out a “survival of the fittest” story where only women who do not need amenities gain enough experience to become more productive, which would not be consistent with adverse selection. Finally, I provide evidence that this adverse selection issue also exists among the pool of women who might be attracted to the firm by the provision of amenities. I predict productivity among an “applicant sample” of women who have applied to the firm but not worked there. I am able to predict productivity with a R^2 of 0.2, and find a similarly negative relationship between productivity and WTP in this sample as well.

Why do the least productive women value these amenities the most? Upon examining covariates, high-WTP women

appear more disadvantaged in a variety of ways. For example, WTP for transportation is higher among women who face higher commuting costs, feel less safe on public transport, have lower household income (excluding own), have not worked at all in the past 3 months, and who experience more symptoms of depression and anxiety. Women valuing childcare more come from lower-income and households with more conservative gender norms, and women valuing more frequent paychecks come from households with more precarious finances. Together with the negative correlation with productivity, these facts suggest that the women who need higher quality jobs the most are also the women the firm would be the least keen to hire if they could observe productivity *ex ante*.

While I have demonstrated that shifting compensation towards amenities instead of wages would induce adverse selection, I also document large WTP. To the extent that offering job amenities might dramatically increase labor supply, especially in this context where female labor supply is constrained by lack of access to transportation and childcare, the net effect on firm profit remains ambiguous. To determine whether the labor supply increasing mechanism could outweigh the adverse selection mechanism, I write down a monopsony model with two key additions: 1) the firm may use wages and amenities to attract workers and 2) the average productivity of workers is allowed to depend on the wage/amenity levels. In preliminary results, I find that mandating job amenities while allowing the firm to lower wages to the profit-maximizing level would result in lower profits. The mandate also results in lower average worker productivity via selection but would increase firm employment substantially. The increase in firm employment would disproportionately benefit disadvantaged women who need these amenities the most.

The government of Jordan is interested in increasing female labor market access and supported the funding of this research to learn about cost effective policies to achieve this goal. My research shows that firms that shift compensation towards female-friendly amenities will likely experience lower profits, at least in the short-medium term. This suggests that simple nudges aimed at firms to increase amenities will likely not succeed. Policy makers should instead consider alternative solutions such as subsidies or direct provision of childcare, transportation, etc as a more effective means of promoting women's employment.

Section 2 describes the setting, with context on Jordan, the firm, and sample of enumerators. Section 3 describes the survey data, including the WTP data on job amenities, and the administrative productivity data. Section 5 describes the empirical strategy for estimating the relationship between WTP and productivity, and section 6 reports the results. Section ?? writes down a monopsony model with job amenities and adverse selection to interpret the results found in the previous section. Section ?? fits the model to data on worker preferences and productivity to discern how different counterfactual policies like an amenity mandate might impacts outcomes like firm profit, employment, wages and worker utility.

2 Setting

2.1 Jordan

Jordan is located in the Middle East and bordered by Saudi Arabia, Syria, Iraq, Israel, and the West Bank. It is politically stable and classified as lower-middle income by the World Bank. It experiences relatively high rates of unemployment (ranging from 10 to 20 percent), and 35 percent of employment is in the informal sector. Approximately half of the population has completed secondary school, and 20 percent have some kind of post-secondary education. The gender gap in education has shrunk dramatically in the past several generations, with women under 40 currently attaining more education than their male counterparts.

Despite near-equal rates of education between the sexes and gender norms that are comparable to, if not more liberal than peer countries in the MENA region, Jordan ranks in the bottom third in the region with respect to the ratio of female/male labor force participation, and ranks 6th lowest in the world in female labor force participation (World Bank, 2025; ILO, 2025). While cultural norms contribute to these low rates of employment, logistical barriers also play a large role. Approximately 40 percent of Jordanian respondents (male and female) reported in the 2020-2021 wave of the Arab Barometer that the biggest barrier to women entering the workplace in their country was lack of access to childcare or transportation (higher than the regional average of 34 percent), and an additional 24 percent cited low wages. Less than a quarter cited a cultural reason, such as working being socially unacceptable. In fact, female respondents were far more likely to cite structural instead of cultural reasons as barriers to women's employment than men: 74 percent of female respondents in Jordan cited lack of transportation, childcare or low wages as the biggest barrier. For these reasons, Jordan is a particularly salient context in which to study how job amenities impact labor supply and worker selection, especially among women.

2.2 Research Firm

The research firm I partnered with is one of the largest and most professional research firms in Jordan, and handles many of the large surveys conducted by the government, UN agencies, and academics. The firm employs 30-40 full time formal staff who manage projects, data, finances, human resources, etc. The firm also hires hundreds of temporary informal workers to work as survey enumerators, drivers, and other staff who interact with respondents to schedule and conduct interviews. These survey enumerators comprise the research sample.

In most developing countries, using survey enumerators to collect data from respondents remains the norm. In general, survey enumerators typically have higher levels of education than average so that they are able to help respondents with lower levels of education or understanding respond accurately to questionnaires. In this setting, many female respondents would feel uncomfortable sitting alone with a male enumerator, and any woman at home

during the survey would need to dress appropriately for a male guest, so nearly all enumerators are women.

At the start of each project, all enumerators are trained on how to administer the questionnaire and other relevant tasks like using the tablet, gaining consent ethically, etc. Phone surveys are either conducted at the enumerator's home or at the central office in Amman, depending on the project. For projects with in-person surveys, the firm recruits enumerators that live in the vicinity of respondents. Enumerators are responsible for traveling to a central meeting point, from which a logistical coordinator drives 2-3 enumerators to respondent's homes.¹ Because addresses are often not precise, the coordinator usually communicates with the respondent to locate the home. The coordinator will also be in touch with the project manager throughout the workday as needed. The enumerator then knocks on the respondent's door, gains consent, and conducts the survey inside the respondent's home or nearby. Introducing oneself and the survey succinctly before gaining consent requires considerable social and communication skills. Administering the surveys quickly and accurately requires attention to detail, good communication skills, patience, and a strong working memory, in addition to sufficient understanding of the meaning of survey questions. Sensitive or long surveys also benefit from strong social skills on behalf of the enumerator.

When staffing for new projects, the firm prefers rehires, for whom they have observed productivity information, but conducts new hiring as needed. After each project, enumerators are scored on a scale of 0-5 on their performance, and the firm keeps a database where each enumerator's average assessment score can be referenced. When staffing projects with rehires, the project manager decides on an appropriate score threshold based on the difficulty or sensitivity of the survey, and recruits experienced enumerators with scores above that threshold. Conditional on clearing the threshold and fitting any geographic requirement, the firm contacts these workers arbitrarily. To recruit new hires, the firm posts advertisements in public places with a QR code that links to a 5-minute online application. The application requests basic identifying information, educational attainment, and whether the applicant has any prior experience as an enumerator. It also requires applicants to watch several short videos with more information about the job and working conditions. The firm requires at least a post-secondary degree and prior experience is favorable, but there is no additional screening such as an interview. As such, productivity can be considered unobserved at the time of hiring.

2.3 Amenities

Willingness to pay was elicited for 4 amenities: transportation to and from home instead of a central meeting point for field projects, a childcare voucher for the duration of the project to be used at a center of the worker's choosing, monthly paychecks instead of pay at the end of the project, and fixed daily pay instead of piece rate pay. These amenities were selected to be relevant to the specific job, desired by workers, and feasible to provide to be incentive compatible.

¹In the case of random block sampling, the coordinator drives the enumerator to the correct location and the enumerator knocks on doors on foot to invite potential participants to the survey.

I conducted qualitative interviews with existing workers with the help of a translator unaffiliated with the firm so that workers would feel free to speak freely. Interviewees were asked about their workdays from start to finish including their transportation and care arrangements, the most challenging and most rewarding parts of those days, what motivated them to seek employment and why they took this job in particular, and what benefits or amenities would improve their experience the most.

Transportation was identified as a major issue, especially for workers living in rural areas, and there was broad interest in childcare among mothers with young children. Workers also expressed that waiting for pay until a project ended is burdensome especially when the household had urgent financial needs, and there was agreement that monthly paychecks were preferred since this is the standard paycheck frequency in Jordan.² Workers also disliked being paid per survey completed since it makes pay unpredictable. While performance is related to effort and skill, sometimes respondents simply do not want to consent to be surveyed, as is their right, or do not pick up the phone, etc.

Other amenities workers were interested in but were not chosen for the study include access to social security, permanent employment instead of project-based contracts, and the ability to work from home.³ The former two were not included since it was not possible to provide these amenities for the sake of this project and therefore not possible to elicit WTP in an incentive compatible manner. While the firm does often allow workers doing phone surveys to work from home, the experience of conducting a phone survey differs from an in-person survey in many other ways besides the work-from-home aspect. For example, many of the workers interviewed actually preferred field projects since it is much easier to hold a respondent's attention in person than over the phone. This means that comparing a phone survey job to a field project job would not isolate the work-from-home amenity value.

2.4 Sample

Sample Composition: Towards studying adverse selection, the survey was conducted on a sample of individuals who had ever worked for the firm as enumerators ("workers") and individuals who have applied to the firm to work as enumerators, met the education requirements, but have not worked ("applicants"). Workers have observed productivity which is critical to the analysis, and applicants well approximate the pool of potential workers who are marginal to the job and may be affected by changes to wages or amenity policies.

There are two types of people within the applicant pool: those the firm did not (yet) call to offer a job, and those the firm called who declined. While I do not have data on who has been contacted directly, I do have information on whether individuals are "active" or "inactive". To become inactive, one must signal their disinterest in the job or repeatedly decline the firm's calls. Therefore, those who have applied but not worked and are still active have likely

²The firm's practice of paying enumerators after data collection is complete is likely related to the timelines with which grants are disbursed.

³Other challenges inherent to this occupation include a lack of access to restroom facilities in the field, extreme temperatures especially while walking door-to-door in the summer, and the emotional toll of conducting sensitive surveys about food insecurity, trauma from forced displacement, etc.

not yet been contacted, while those who have not worked but are inactive must have declined a job. Because the firm does not conduct productivity screening like interviews and contacts applicants for work arbitrarily conditional on geographic fit, the only systematic differences between those who have been contacted but declined and those who have not been contacted should be geographic. Indeed, the difference between columns 2 and 3 in Table 2 reveals that the only difference between those who declined and those who have not been contacted is that the firm contacts workers from Amman and urban areas more often. This is consistent with the fact that the firm has a harder time recruiting in Amman (possibly since there are more outside options in Amman).

Comparing workers to applicants who declined a job offer will be informative on selection into working. Those who decline are slightly younger, have higher levels of education, are more likely to live in Amman, are more career oriented and live in households which are less financially insecure. These facts suggest that applicants who decline are declining because their reservation wage or utility is higher than that which is provided by this job.

External Validity: While this sample is not formally representative, it is an excellent sample of interest to study how the provision of job amenities may affect worker selection and labor supply. The job application was short, simple, and posted in public places in order to be seen by women open to work but potentially not actively searching. As this job has low barriers to entry in terms of application and hiring, it well approximates the type of job available to a woman who is marginal to the labor force.

Table 1 reports demographic information for the study sample and the most recent labor force survey. The women in this study are of similar age and marital status as the average woman in the labor force in Jordan. The women in the study sample are more educated than the population due to job requirements, but the difference in educational attainment between the study sample and the population of women in the labor force is much smaller than the population at large. This paper’s sample is slightly more rural than the population, and has rates of labor force attachment which are lower than the average woman in the labor force but higher than the average woman.

3 Data

This paper combines two key datasets: a survey of workers and applicants measuring willingness to pay for job amenities and other covariates, and administrative firm data including application and productivity data. I discuss each in turn.

3.1 Firm Data

The firm provided several datasets related to worker productivity as well as deidentified information from the application database. The productivity data fall primarily in two categories: supervisor evaluations and survey metadata.

Supervisor evaluations: After each project, each enumerator receives a score on their performance from the Operations team and from the Data Processing team. Operations is in charge of day-to-day management of enumerators and provides a score based on the enumerator’s professionalism, perceived competence, the number of surveys conducted, etc. Data Processing assigns a score based on data quality assessed during the data cleaning process. The firm maintains a database of these scores and consults the enumerator’s average evaluation when making future hiring decisions.

Enumerators whose performance falls well below expectations or who commit fraud or other unethical behavior are added to a “blacklist” and will not be hired again. The firm also has a policy of adding enumerators who are extremely tardy to training sessions to the blacklist as a high-powered incentive to promote punctuality. Based on analysis triangulating the dates on which individuals were added to the blacklist against the dates of projects they worked on, it appears that at least half of individuals were added to the blacklist due to tardiness, not poor performance.

Survey Metadata: The firm’s standard practice is to code the client’s survey and host the survey on their own CTO server.⁴ The firm prepared metadata from each survey stored on their server dating back to 2020. Each observation from this dataset corresponds to a survey completed, and contains non-identifying information about that survey, including the enumerator who conducted it, the project it was associated with, the date and time the survey was conducted, the duration in seconds, the callback accuracy where available,⁵ and the share of questions which were answered with "Don’t Know", "Refused", or "Other".

This dataset also includes the score the enumerator earns on the project’s training exam, as well as information about the project including whether it was phone-based or in the field and whether it was paid per day or per survey.

Applicant Database: The firm also shared the deidentified applicant database, which includes information on age, governorate of residence, educational attainment, prior experience, and employment preferences. This information is available for applicants and workers.

3.2 Survey Data

The survey was conducted over the phone during the summer of 2024. Workers and applicants were invited to participate in randomized order until the total number of surveys was completed. Enumerators conducting the survey were employed by the same firm but excluded from the sample.

⁴Clients may elect to host surveys on their own CTO server. For example, this project was hosted on a private server to ensure the firm did not have access to identified worker surveys.

⁵The firm’s standard practice is to call back approximately 10 percent of respondents within a week or so of the survey to ask them 5-10 short questions from the survey whose responses should not have changed (ie, birth date, educational attainment, etc.) to assess

3.2.1 Elicitation of Willingness to Pay for Job Amenities

The first 4 sections of the survey measured willingness to pay for 4 job amenities separately in randomized order: transportation, childcare, monthly paychecks instead of lump sum payment at the end of the project, and daily wages instead of piece rates. Each section implemented a series of discrete choice questions akin to a multiple price list where the respondent is asked to choose between job A, which pays 15 JD per day and includes the amenity, or job B, which pays x JD per day and does not include the amenity, where $x \in \{15, 30\}$. The respondent is told that the jobs are otherwise identical. The wage of job B is varied until the respondent switches from job A to job B (or vice versa if the price list is descending), revealing her WTP.

The consent script explained that participants would receive a small incentive for participation (3 Jordanian Dinar) and would also be entered into a random draw to receive the job (A or B) that they chose in a randomly selected question if they were selected by the firm to work again. The high take-up rate of a relatively long phone survey suggests that the incentivization was well understood.

At the start of each amenity section, several sentences of relevant details about the job and the amenity were provided. The transportation, childcare, and paychecks sections described the project as lasting for two months and having a daily target of 5 surveys. Two months was selected to be long enough to be worth adjusting one's schedule for, and to be a reasonable comparison for lump sum vs monthly pay. 5 surveys per day connotes a survey that is of medium length and complexity. Childcare was described as a voucher to be used at a childcare center of their choosing for the duration of the project, and transportation was described as a trusted driver affiliated with the firm driving them to/from home instead of a meeting point. Monthly paychecks was described as guaranteed monthly paychecks at the end of each month instead of a lump sum payment at the end of the project, the status quo. For the daily pay amenity, the project was described as lasting 2 months and having a target of 3 surveys per day with extra information like the sampling method, expected range of survey duration, and the transport time between respondents to help the respondent gauge her expected productivity accurately.⁶ The full text of the survey instrument is available in appendix XX. Attention questions were included to ensure the respondent internalized the relevant details of the job. Between 85-90 percent of respondents answered these questions correctly the first time.

Jack, McDermott, and Sautmann (2022) describes various pitfalls that may arise during a multiple price list elicitation of WTP that introduce bias. Care was taken in the design of the multiple price list to avoid these sources of bias while remaining practical for elicitation over the phone.

Order Effects: To avoid order effects, both the order of sections eliciting WTP for the 4 amenities and whether each price list was ascending or descending were randomized.

Adaptive Bias: Adaptive surveys may introduce bias when combined with respondents who make mistakes or change

⁶These facts were based on a real survey previously conducted by the firm for which I have the data.

their mind halfway through the elicitation. For example, if the respondent accidentally misreports that she prefers job B in a choice between A paying 15JD + transport and B paying 20JD, an adaptive survey will anchor on this amount and narrow in on more precise WTP around this incorrect (too small) value. To avoid this problem while remaining minimally tedious, the survey first asked 4 questions, increasing or decreasing the pay of job B in increments of 5 dinar with no adaptiveness. For example:

- If job A pays 15 JD/day and includes childcare, and job B pays **15** JD/day without childcare, which would you prefer?
- If job A pays 15 JD/day and includes childcare, and job B pays **20** JD/day without childcare, which would you prefer?
- If job A pays 15 JD/day and includes childcare, and job B pays **25** JD/day without childcare, which would you prefer?
- If job A pays 15 JD/day and includes childcare, and job B pays **30** JD/day without childcare, which would you prefer?

Assuming there are no logical inconsistencies in her choices, the survey then asks questions adaptively in 1JD increments until her WTP is determined within a 1 JD range. For example:

- If job A pays 15 JD/day and includes childcare, and job B pays **15** JD/day without childcare, which would you prefer? **A**
- If job A pays 15 JD/day and includes childcare, and job B pays **20** JD/day without childcare, which would you prefer? **A**
- If job A pays 15 JD/day and includes childcare, and job B pays **25** JD/day without childcare, which would you prefer? **B**
- If job A pays 15 JD/day and includes childcare, and job B pays **30** JD/day without childcare, which would you prefer? **B**
- Begin adaptive section:
- If job A pays 15 JD/day and includes childcare, and job B pays **23** JD/day without childcare, which would you prefer? **B**
- If job A pays 15 JD/day and includes childcare, and job B pays **22** JD/day without childcare, which would you prefer? **A**

At the end of the elicitation, the enumerator confirms with the respondent her choices to make sure it aligns with her preferences and that no mistakes were made. “So, when job B pays between 15-22 JD you preferred job A (15 JD with transportation), but when job B pays 23 or more JD, you prefer job B. Is that correct?”

Multiple Switching: In the event of a logical inconsistency or “multiple switching” from A to B then back to A, the survey prompts the enumerator to explain to the respondent that their choice is logically inconsistent. For example, suppose a respondent made the following choices:

- 15 JD + transportation $\succ$ 15 JD
- 15 JD + transportation $\prec$ 20 JD
- 15 JD + transportation $\succ$ 25 JD
- 15 JD + transportation $\prec$ 30 JD

The survey would read: : “You preferred job B over job A when job B paid 20JD, but then you said you preferred job A over job B when job B paid 25JD. Are you sure about this?” Enumerators were trained to explain to the respondent why this was illogical if needed.

Never Switching: Open ended questions were asked if respondents never switched their choice. If they always preferred job B (implying that they value the amenity at zero), the enumerator asked them to explain why they always preferred job B, and these responses were scrutinized after data collection ended for logical coherency.⁷ Respondents who always

⁷Reasons for not valuing childcare include having family members available for caregiving; reasons for not valuing transportation

preferred job A (ie, WTP is larger than the range of the price list) were asked to report their true switch point with the following language: “What would job B’s wage need to be to persuade you to switch from A to B? Your answer will not affect the random draws in any way.”⁸

3.2.2 Covariates

Following the WTP sections, the phone survey included the following modules: childcare arrangement and attitudes towards different childcare arrangements, commuting arrangements, basic time use in the past 7 days, employment history and attitudes/motivations, household income, savings/loans, gender attitudes. Two additional modules were included to predict productivity: a digit span test of working memory and an Arabic-validated Big 5 personality test. Enumeration requires listening to respondent’s responses attentively and remembering/recording accurately, which is related to working memory. Successfully gaining consent from cold-call or door-knock respondents requires high levels of extroversion and agreeableness, and conscientiousness is likely correlated with careful data collection.

4 Descriptive Statistics

Willingness to pay for all of the amenities is on average quite high. Figure 9 reports the average willingness to pay reported in percentage terms of the base wage separately for workers, applicants who turned down a job, and those not yet contacted. Concretely, a respondent who preferred job A (pays 15 JD + amenity) when job B paid 19 but then switched to job B when job B paid 20 JD has a WTP between 4-5 JD. Taking the midpoint, their WTP is 4.5, which is 30% of the base wage. The average worker is willing to forgo a 55 percent increase to their wages in order to work in a job that includes transportation, 42 percent for childcare (among mothers), 35 percent for monthly paychecks and 30 percent for daily wages. These means are very high, but conceal a great deal of variation across person, displayed in the histograms in Figure 3. Childcare and paychecks have distributions which are fairly bimodal, consistent with 10-20 percent of workers having free care from family or no liquidity constraints respectively, while another 10-20 percent of the population are so constrained by these issues that they prefer the amenity to a doubling in wages. WTP for transportation has more variation across the distribution but also features a thick right tail. 8 percent of workers prefer transportation to an increase to wages by 150 percent. Finally, most workers prefer daily wages to piece rates, but approximately 5 percent are indifferent, and nearly 30 percent have some degree of preference for performance pay.

The other stylized fact revealed by the summary statistics on WTP is that applicants have higher WTP than workers for all amenities. The difference between WTP as a percent of base pay between workers and applicants

include having her own car, or enjoying a short walk to work, etc. Reasons for not valuing monthly paychecks often mentioned reasons related to saving/self-control.

⁸While this language does not directly state “what is the *lowest* wage that would prompt you to switch”, it was found to be implied in piloting, and that “lowest” introduced confusion.

ranges between 20-30 percentage points. Additionally, applicants who turned down a job have either higher or statistically indistinguishable WTP for all amenities than those not yet contacted. This is consistent with having turned down the job for reasons related to their higher taste for these amenities which are not currently offered.⁹

Table 5 reports covariates which predict increased WTP for each of the amenities, controlling for whether or not the respondent is worked for the firm and whether or not they've been contacted and governorate fixed effects. Many of these relationships are sensible: for example, higher commute costs are associated with higher WTP for transportation, increased financial strain is associated with higher WTP for regular paychecks, and the children's grandmother being unavailable for childcare increases WTP for the childcare amenity.

WTP for transportation is higher among rural women, women with higher commute costs, and those who feel unsafe on public transportation more often. The safety relationship is statistically and economically significant, though it should be noted that public transportation is relatively safe for women in Jordan. 63 percent of respondents reported always feeling safe on public transportation, so this effect is driven by a minority of the population. WTP for transportation is also 1.3 JD (11 percent) higher among women who report that transportation is the largest barrier preventing women in Jordan from entering the labor force. Somewhat surprisingly, the length of respondents' commutes to their job with the firm was uncorrelated with WTP. Finally, income is weakly negatively correlated with WTP for transportation, meaning that firm-provided transportation was more attractive to lower income women.

On the other hand, WTP for childcare is uncorrelated with household income, how much women currently pay for daycare (if using), or even how close the nearest acceptable daycare is to their home. Somewhat mechanically, being willing to use paid childcare, citing childcare as the primary barrier to women's labor force entry, and the respondent's mother or mother-in-law being unavailable for childcare are highly correlated with WTP. Demand is also higher among women with more conservative gender attitudes and women whose male family members disparage women who work, suggesting that firm-provided childcare could make working outside the home more palatable for more conservative households. WTP is also higher among women who are currently working but want a different job in the next two years (either different employer or different job within the same employer), and women who have not worked in the past 3 months. Taken together, these facts show that demand for childcare is low among the 20-30 percent of mothers who don't need or want it, but extremely high among the rest for whom firm-provided childcare could ease frictions which prevent them from working in the capacity they desire. Indeed, over 75 percent of all mothers with young children reported they would work longer hours, open a business, or begin working again if they had access to childcare even for a fee.

WTP for monthly paychecks instead of lump sum payments is higher among those with little savings and no one from whom they can borrow money easily. It is also higher among those who reported they had to reduce expenditure or

⁹It is possible that lived experience with the job directly impacts willingness to pay for the amenities, although its not clear why experience with the no-amenity job would decrease taste for the amenities. It seems equally plausible that the lived experience of taking 2 hours of buses might increase WTP for transportation, for example.

take out a formal loan the last time their paycheck was late, and those who have a formal loan.

The main predictors of willingness to pay for fixed wages include measures of labor force attachment, experience with the firm, and the motivations for work. Women with more labor attachment and more attachment to the firm are less averse to performance pay, while those motivated primarily by earning money are *more* averse to performance pay, presumably because it increases variability in income.

It should also be noted that a higher score on the Big 5 neuroticism scale is associated with higher WTP for childcare and regular monthly paychecks, and the point estimate for transportation is similarly large with a p-value of 0.106. An Arabic-validated version of the questionnaire was used, and many of the questions pick up symptoms of depression and anxiety ("I feel sad", "I feel anxious", "I am a lonely person", etc). This association is not mediated by any obvious demographic characteristic such as income, location, age of children, education, etc. This finding suggests that individuals prone to negative emotions and/or mental health challenges particularly value amenities which make work less stressful.¹⁰ It should be noted however, that these (and all the personality measures) do not mediate the relationship between productivity and WTP. More detail is discussed below.

5 Empirical Strategy

To measure the relationship between productivity and WTP for amenities, I regress stacked measures of productivity (θ_{ijk}) on individual-level willingness to pay (WTP_i , clustering at the individual level. Since an individual's true productivity is measured with error, putting it on the left hand side ensures that the estimates are not affected by attenuation bias.

$$\theta_{ijk} = \beta_0 + \beta_1 WTP_i + \epsilon_{ijk} \quad (1)$$

θ_{ijk} is an enumerator-outcome level measure of productivity that stacks all available measures of productivity for an individual. The preferred specification includes the following: project-level standardized measures of the number of surveys per day, total number of surveys completed in the project, callback accuracy, and training exam score; year-level averages of the enumerator's operations and data processing evaluations; and their overall assessment score. These measures are preferred since the directionality is clear: higher accuracy is always better, higher evaluations are always better etc. In robustness two additional outcomes are included: survey duration and inclusion on the blacklist (inverted as appropriate). While shorter surveys are on average indicative of skill, particularly short surveys can indicate fraud or cut corners. Because one can be added to the blacklist for being late to a single training session or for committing grossly unethical fraud against respondents, the best way to parameterize it is unclear.¹¹

¹⁰Is that why I ended up studying this? Who can say?

¹¹Additionally, the other measures of productivity will generally capture an enumerator's poor performance which leads to being added

The preferred specification uses the version of WTP which censors WTP at the top of the price list for individuals who preferred job A (lower wage + amenity) for every wage in the price list. In robustness, the uncensored version of WTP that uses a free response for individuals whose WTP was outside the bounds of the price list is used. The results are qualitatively similar.

6 Results

Reporting the raw data, figure 2 shows binscatters of WTP for each amenity against the productivity index. Each panel shows a downwards sloping line of best fit, indicating a negative correlation between WTP and the productivity index. The tables that follow display this negative correlation in numeric form and investigate the extent to which this correlation can be explained using information observable to the firm or information observable to the researcher.

Table 6 reports the regression coefficients from equation 1. WTP for transportation, monthly paychecks, and fixed wages are all statistically significantly negatively correlated with productivity. Column 1 reports that a 1 JD increase in WTP for transportation is associated with -0.007 SDs of productivity. The SD of censored WTP for transportation is 8.1, implying that a 1 SD increase in WTP is associated with 0.006 SD lower productivity. WTP for monthly pay is associated with .006 SDs lower productivity (or a 1 SD increase in WTP is associated with a 0.033 SD decrease in productivity). The negative correlation between productivity and WTP for fixed wages is slightly smaller in terms of dinars but similar to WTP for monthly paychecks when scaled by the standard deviation (0.030). Notably, There is no correlation between WTP for childcare and productivity. The sample size for these regressions is smaller since WTP for childcare was only elicited for women with young children making the estimates less precise, but the point estimates are also closer to zero.

Column 5 shows that WTP for fixed pay is negatively correlated with productivity as expected, though becomes insignificant with the addition of location and education controls.

6.1 Robustness

This section examines whether or not the negative association seen in the data is consistent with adverse selection. For that to be true, the negative correlation must not be mediated by something which is observable to the firm, and it must be present ex ante, not caused by differential attrition out of the workforce due to reasons correlated with their preference for amenities.

The first columns of tables 7 and 8 add district and education fixed effects, which are the two main observables to the firm. The negative correlation observed for WTP for transport and monthly pay is robust to these controls, to the blacklist. Very low achievement will appear in the survey counts, making up responses will appear in the callback accuracy, inappropriate conduct will appear in the evaluations, etc.

while the negative correlation between WTP for fixed pay becomes indistinguishable from zero (though still negative) after adding these fixed effects. This means that some of the signalling variation in WTP for fixed pay falls along demographic lines that are observable to the firm, while the information on a worker’s expected productivity contained in her WTP for transportation or monthly pay is not otherwise observable to the firm. Columns 3 and 4 of tables 7 and 8 show that these findings are robust to other productivity specifications.

Columns 5 and 6 add fixed effects for the worker’s n ’th project with the firm to provide evidence that the negative correlation is present *ex ante* and not the result of experience effects. If the correlation is robust to this control, then it rules out a “survival of the fittest” mechanism, where workers who value the amenity the most quit earlier and appear less productive in the cross section due to returns to experience. It also rules out any possibility that the negative correlation is a result of experience causing both increased productivity and decreased taste for the amenity. Data on the worker’s n ’th project is only available in the survey metadata productivity dataset. The negative point estimate remains statistically significant, showing that even after holding location, education, and experience fixed, the worker who values transportation or regular pay more will be less productive on average. This is strong evidence that a firm that lowered wages in exchange for these amenities would likely attract less productive workers.

What does mediate the relationship between productivity and WTP for transportation and monthly pay then? Table 9 makes use of the detailed covariates collected in the phone survey to investigate whether adding more rich controls than the firm has access to can explain the negative correlation. Column 1 adds controls for basic demographics (marital status, number of children, age, etc), household income, personality test scores, and gender attitudes. These controls attenuate the relationship between productivity and WTP for transport but it remains statistically significant. Column 2 adds all controls collected on the type of commute, duration, cost, safety, etc, and the estimate remains statistically significant. This finding suggests that the true mediator is less related to commuting logistics and more related to preferences for convenience.

How does this finding square with the fact that WTP for transportation is positively correlated with commute cost and lack of safety? In short, because the correlation between commute aspects and productivity and commute aspects and WTP are both well below 1. The following (true) long and short regressions are presented to demonstrate the omitted variable bias logic, where S_i is self reported safety on public transport:

$$\begin{aligned}\theta_{ijk} &= \alpha - .00654 WTP_i + 0.0327 S_i + \epsilon_{ijk} \\ \theta_{ijk} &= \alpha^* - .00688 WTP_i + \varepsilon_{ijk} \\ S_i &= -.0101 WTP_i + \eta_{ijk} \\ &\rightarrow -0.0068 = -0.00654 + (0.0327) * (-.0101)\end{aligned}$$

This finding also makes it implausible that the provision of transportation might have a heterogeneous treatment effect on less productive women that cancels out the adverse selection mechanism. If the negative correlation were entirely driven by exhaustion from a long commute, for example, then any negative correlation between WTP and productivity would disappear once the amenity was provided since it was the lack of the amenity which was causing them to be less productive in the first place. The fact that controlling for income, commute time, mode of transport, cost, etc does not eliminate the effect suggests that, were transportation provided and the wage lowered, the average worker attracted to the job would indeed be slightly less productive.

A similar exercise for monthly paychecks reveals that the only predictive mediator is financial precarity, defined by the combination of not having approximately 330 USD PPP in savings and having no one from which they could borrow money in an emergency. Interestingly, income alone does not do anything to explain the correlation. Because controlling for financial precarity does attenuate the effect, it's more plausible that more regular paychecks might have productivity-enhancing effects *à la* Kaur et al., 2024. Because this job is inherently temporary, it seems equally plausible however that this single amenity is not able to address whatever structural reason results in both lower productivity and financial precarity.

In conclusion, less productive workers have stronger preferences for transportation and monthly paychecks for reasons that are almost entirely unobservable to the firm. If the firm were to lower wages in order to provide these amenities the average productivity of workers would likely fall. However, section 4 also shows that women have extremely high willingness to pay for amenities, especially transportation. To the extent that women's utility from working is continuous and that there are women on the margin of working and not working, these very large WTP imply that offering amenities like transportation could dramatically increase labor supply available to the firm. Depending on how large the labor supply response is, offering amenities could increase profit enough to outweigh the adverse selection mechanism. To estimate the net effect offering amenities would have on firm profit, total employment, and worker utility, I next turn to a monopsony model.

7 Case Study: Counterfactual Policies on Firm-Provided Transportation

The foregoing sections have shown that women in this setting have very large WTP for firm-provided transportation in particular, and also that there is considerable evidence that offering this amenity would induce adverse selection in worker productivity. To estimate the net effect of offering transportation to workers, I write down a monopsony model where the firm may use wages and/or amenities to grow large. Workers have outside options distributed according to a shifted Beta distribution and accept employment if the wage/amenity package exceeds their outside option. Solving the firm's profit first order condition with respect to wages generates a profit-maximizing wage with and without the amenity. I use knowledge of the firm's current profit-maximizing wage without transportation to back

out the correct rescaling of productivity in standard deviations to dinars. I then examine various counterfactuals, including mandating or subsidizing transportation.

7.1 Workers

Let an individual's utility from accepting an offer from the firm be given by

$$U_i = \ln(w) + v_i a - \ln(b_i)$$

where w is the offered wage, a is an indicator for the amenity, and v_i is individual i 's taste for the amenity and b_i is the worker's outside option. Individuals accept the offer if $U_i > 0$. Using this functional form, v_i can be written as a function of the reference wage and WTP : $\ln(w) + v_i = \ln(w + WTP_i)$. Introducing curvature in wage ensures that a person who prefers the amenity over doubled wages does not prefer working for free with the amenity to the reference wage.

Define $\tilde{b}_i := \frac{\ln(b_i) - \ln(\underline{b})}{\ln(\bar{b}) - \ln(\underline{b})}$ and assume $\tilde{b} \sim \text{Beta}(\phi, 1)$.¹² Then the share of the population accepting the job at wage w and amenity a , $F(w, a)$, can be modeled as $\text{Prob}(U_i > 0)$. $F(w, a)$ is a key input into the firm's profit function.

$$\begin{aligned} F(w, a) &= \text{Prob}(U_i > 0) \\ &= \text{Prob}(\ln(w) + v_i a - \ln(b_i) > 0) \\ &= \text{Prob}(\ln(w) + v_i a > \ln(b_i)) \\ &= \text{Prob}\left(\frac{\ln(w) + v_i a - \ln(\underline{b})}{\ln(\bar{b}) - \ln(\underline{b})} > \frac{\ln(b_i) - \ln(\underline{b})}{\ln(\bar{b}) - \ln(\underline{b})}\right) \\ &= \text{Prob}\left(\frac{\ln(w) + v_i a - \ln(\underline{b})}{\ln(\bar{b}) - \ln(\underline{b})} > \tilde{b}\right) \\ &= \left[\frac{\ln(w) + v_i a - \ln(\underline{b})}{\ln(\bar{b}) - \ln(\underline{b})} \right]^\phi \end{aligned} \tag{2}$$

Total utility to workers generated by this firm is given by expected value of utility above the outside option for those who chose to work. This quantity is useful for assessment welfare impacts of different policies since it accounts for both employment effects and direct utility from wages/amenities. In a slight abuse of notation, denote $\widetilde{\ln(w) + v_i a} :=$

¹²This assumption is similar to the assumption made in Kline (2025), section 4.2. The main deviation from Kline (2025) is that wages are log, not linear. Linear wages allow for a closed form solution but generate nontrivial amounts of employment at zero wages in this sample if transportation is included, which is not plausible in this setting.

$\frac{\ln(w)+v_i a - \ln(\underline{b})}{\ln(\bar{b}) - \ln(\underline{b})}$, which rescales $\ln(w) + v_i a$ into $[0,1]$ $\tilde{b}$ space.

$$\begin{aligned}
U &= \int_{v=-\infty}^{\infty} \int_{\ln(\underline{b})=-\infty}^{\ln(w)+v_i a} (\ln w + v_i - \ln(\underline{b})) f(v, \ln(\underline{b})) dv d\tilde{b} \\
&= \int_{v=-\infty}^{\infty} \ln(w) + v_i \int_{\ln(\underline{b})=-\infty}^{\ln(w)+v_i a} -\ln(\underline{b}) f(v, \underline{b}) dv d\tilde{b} \\
&= \int_{v=-\infty}^{\infty} (\ln(w) + v_i) f(v) \int_{\tilde{b}=0}^{\widetilde{\ln(w)+v_i a}} (-\tilde{b}(\ln(\bar{b}) - \ln(\underline{b})) + \ln(\underline{b})) \phi \tilde{b}^{\phi-1} dv d\tilde{b} \\
&= \int_{v=-\infty}^{\infty} (\ln w + v_i) f(v) \int_{\tilde{b}=0}^{\widetilde{\ln(w)+v_i a}} -(\ln(\bar{b}) - \ln(\underline{b})) \phi \tilde{b}^{\phi} - \ln(\underline{b}) \phi \tilde{b}^{\phi-1} dv d\tilde{b} \\
&= \int_{v=-\infty}^{\infty} (\ln w + v_i) f(v) \left[-(\ln(\bar{b}) - \ln(\underline{b})) \frac{\phi+1}{\phi} [\widetilde{\ln(w) + v_i a}]^{\phi+1} - \ln(\underline{b}) [\widetilde{\ln(w) + v_i a}]^{\phi} \right] dv
\end{aligned}$$

Finally, workers in the model have time invariant productivity p_i . For tractability, assume outside options are orthogonal to the joint distribution of v_i and p_i .¹³

7.2 Firm

Assume the firm is a monopsonist which decides whether to offer the amenity (a) and sets a wage (w) to maximize profits. The firm knows the joint distribution of worker productivity p , amenity valuation v , and outside options b , but cannot observe this information for a given worker ex ante. Then firm profit is given by the expectation of worker product minus wages minus amenity cost (if provided) over workers willing to work at wage w and amenity level a . Firm profit is increased by hiring more productive workers, lowering the wage/amenity bill, or increasing total employment (since the firm makes positive profit on each worker).

$$\begin{aligned}
\pi &= \int_{p=-\infty}^{\infty} \int_{v=-\infty}^{\infty} \int_{\tilde{b}=0}^{\frac{(\ln w + v_i a - \ln(\underline{b}))}{(\ln(\bar{b}) - \ln(\underline{b}))}} (p_i - w - \psi a) f(v, p, \tilde{b}) d\tilde{b} dv dp \\
&= \int_{p=-\infty}^{\infty} \int_{v=-\infty}^{\infty} (p_i - w - \psi a) f(v, p) \int_{\tilde{b}=0}^{\frac{(\ln w + v_i a - \ln(\underline{b}))}{(\ln(\bar{b}) - \ln(\underline{b}))}} f(\tilde{b}) d\tilde{b} dv dp \\
&= \int_{p=-\infty}^{\infty} \int_{v=-\infty}^{\infty} (p_i - w - \psi a) \left[\frac{(\ln w + v_i a - \ln(\underline{b}))}{(\ln(\bar{b}) - \ln(\underline{b}))} \right]^{\phi} f(v, p) dv dp
\end{aligned}$$

Since a is an indicator, the firm determines the profit maximizing wage with and without amenities and offers

¹³If more productive women had better outside options in addition to lower WTP for amenities, it would amplify the wage dispersion high v /low p and low v /high p types. This would also exacerbate the adverse selection mechanism. Because b_i is not observed in the study sample, some simplifying restriction must be imposed.

amenities if doing so generates more profit. Profit-maximizing wage (conditional on a) is given by the FOC with respect to wage.¹⁴

$$\begin{aligned}
\frac{d\pi}{dw} &= \frac{d}{dw} \int_{p=-\infty}^{\infty} \int_{v=-\infty}^{\infty} (p_i - w - \psi a) \left[\frac{(\ln w + v_i * a - \ln(\underline{b}))}{(\ln(\bar{b}) - \ln(\underline{b}))} \right]^{\phi} f(v, p) dv dp \\
&= \int_{p=-\infty}^{\infty} \int_{v=-\infty}^{\infty} \frac{d}{dw} (p_i - w - \psi * a) \left[\frac{(\ln w + v_i * a - \ln(\underline{b}))}{(\ln(\bar{b}) - \ln(\underline{b}))} \right]^{\phi} f(v, p) dv dp \\
&= \left[\frac{1}{(\ln(\bar{b}) - \ln(\underline{b}))} \right]^{\phi} \int_{p=-\infty}^{\infty} \int_{v=-\infty}^{\infty} -(\ln w + v_i * a - \ln(\underline{b}))^{\phi} + \phi(p_i - w - \psi a) \frac{1}{w} (\ln w + v_i * a - \ln(\underline{b}))^{\phi-1} f(v, p) dv dp
\end{aligned}$$

The sample equivalent of this first order condition is as follows.

$$\frac{1}{N} \sum_{i=1}^N \left[-(\ln(w) - v_i - \ln(\underline{b}))^{\phi} + (p_i - w - \psi a) \cdot \frac{\phi}{w} \cdot (\ln(w) - v_i a - \ln(\underline{b}))^{\phi-1} \right] = 0 \quad (3)$$

This first order condition will be estimated using generalized method of moments. To adjust for the fact that p_i is only observed for those who chose to work, inverse probability weights from a regression predicting who chose to work for the firm (conditional on being contacted) will be used.

7.3 Model Estimation

To estimate how wages and employment would be impacted by a mandate that firms provide transportation, I combine estimates of the labor supply elasticity (ϕ) and minimum/maximum outside option ($\underline{b}, \bar{b}$) in this setting with knowledge of the firm's profit maximizing wage without amenities to back out the correct scale of worker productivity in dinars. I then estimate the profit maximizing wages and employment under an amenity mandate.

To estimate the labor supply parameters $\phi, \underline{b}$ and $\bar{b}$, I use nonlinear least squares to fit equation 2. Taking logs on both sides, equation 2 becomes

$$\ln(F(w)) = \phi \ln(\ln(w) - \ln(\underline{b})) - \phi(\ln(\bar{b}) - \ln(\underline{b}))$$

Because it was not possible to measure reservation wages in this study sample, I use data on stated reservation wages among a sample of unemployed women from the Jordan Labor Market Panel Survey which is representative. To ensure comparability, the sample is restricted to Jordanian women who reported being open to data entry, telemarketing or

¹⁴Following Kline et al (2025), the profit-maximizing wage can be shown to be unique since $\frac{d^2}{dw^2} \frac{1}{F(w, a)} > 0$

teaching jobs (92%).¹⁵

The results reported in Table 10 show a minimum outside option of 0.8 JD per day, or approximately 2.66 USD PPP per day. This wage is very low in this setting, but this should be expected of the lower bound. The lowest adjusted full time daily wage reported in the study sample is 1JD, which adds support that this lower bound is reasonable. The estimated maximum outside option is 24.3 JD per day. This is equivalent to the 75th percentile of adjusted full time wages or the 96th percentile of actual monthly wages (many work less than full time). The implied extensive margin labor supply elasticity with respect to wages at the benchmark wage 15 is approximately 2, which is reasonable for this setting. It asymptotes towards 1.75 as the wage approaches the maximum outside option.

With these parameters in hand, plus knowledge of the distribution of earnings from piece rate jobs at the firm, we can use the first order condition without amenities to write down an empirical moment condition that reveals the correct rescaling of productivity from standard deviations into dinars of revenue. Assuming that the empirical productivity index θ_i is well behaved, then p_i should have the same shape as earnings from piece rates (e), scaled by the markdown. In other words, $p_i = (\theta_i \cdot \frac{\mu_p}{\mu_e} \cdot \sigma_e) + \mu_p$, where μ_e and σ_e are the mean and variance of earnings from piece rate work (known), θ_i is the productivity index (known), and μ_p is the mean productivity of workers in dinars (unknown). Taking $\mu_e = w$, estimating equation 3 using GMM reveals the markdown from productivity to wages consistent with the model.

The figure below plots p_i by quartiles of willingness to pay for transportation. Mean productivity is around 20, implying a markdown of approximately 0.67. The figure also displays how groups with higher average WTP are less productive on average: the yellow line plots the productivity of workers in the highest quartile of WTP who generate on average 1 JD fewer in revenue for the firm than workers in the lowest quartile of WTP (blue).

7.4 Counterfactual: Amenity mandate, relax minimum wage

To determine the profit-maximizing wage under an amenity mandate when the firm cannot observe worker characteristics before hiring, I solve equation 3 with amenity $a = 1$ numerically using GMM. Standard errors are bootstrapped to account for uncertainty in ϕ , $\underline{b}$, and $\bar{b}$.

7.5 Counterfactual: Amenity mandate + enforce minimum wage

7.6 Counterfactual: Minimum amenity subsidy to keep firm profit constant

What is the minimum subsidy for transportation that would induce the firm to offer the amenity, assuming the firm can lower wages? To estimate this counterfactual, I introduce a subsidy parameter that reduces ψ by x percent, and

¹⁵The question was only asked of unemployed women.

estimate the subsidy amount that equalizes profit under the amenity and no amenity state of the world.

7.7 Counterfactual: firm offers workers contract choice

Suppose the firm offered workers a choice between a high wage/no amenity and a low wage/amenity contract. This could also take the form of workers being offered the choice to “buy” firm-provided transportation for a certain price. What wages would it set, knowing these wages would attract different types of workers? Could this generate higher utility or profit than offering only one contract?

Let w^h be the high wage offered without the amenity and let w^l be the low wage offered with the amenity. Then there are 5 mutually exclusive types of people: those who would work under either contract but prefer the high wage contract; those who would work under either but prefer the low wage/amenity contract; those who would only supply labor under the high wage contract; those who would only supply labor under the low wage/amenity contract; and those who would not work under either contract.

Prefers high wage contract : $\ln b < \ln(w^l) + v < \ln(w^h)$

Prefers low wage contract : $\ln b < \ln(w^h) < \ln(w^l) + v$

Only works under high wage contract : $\ln(w^l) + v < \ln b < \ln(w^h)$

Only works under low wage/amenity contract : $\ln(w^h) < \ln b < \ln(w^l) + v$

Then profit is the sum of profit generated by these four types of workers:

$$\begin{aligned}
\pi = & \int_{p=-\infty}^{\infty} \int_{v=\ln(b_i)-\ln(w^l)}^{\infty} \int_{\widetilde{\ln(w^h)}}^{\ln(w^l)+v_i} (p_i - w^l - \psi) f(v, p, \tilde{b}) \tilde{b} dv dp && \text{Only works under low wage/amenity} \\
& + \int_{p=-\infty}^{\infty} \int_{v=-\infty}^{\ln(b)-\ln(w^l)} \int_{\widetilde{\ln(w^l)+v}}^{\ln(w^h)} (p_i - w^h) f(v, p, \tilde{b}) \tilde{b} dv dp && \text{Only works under high wage/ no amenity} \\
& + \int_{p=-\infty}^{\infty} \int_{v=\ln(w^h)-\ln(w^l)}^{\infty} \int_{\tilde{b}=0}^{\widetilde{\ln(w^h)}} (p_i - w^l - \psi) f(v, p, \tilde{b}) \tilde{b} dv dp && \text{Prefers low wage/amenity} \\
& + \int_{p=-\infty}^{\infty} \int_{v=-\infty}^{\ln(w^h)-\ln(w^l)} \int_{\tilde{b}=0}^{\widetilde{\ln(w^l)+v}} (p_i - w^h) f(v, p, \tilde{b}) \tilde{b} dv dp && \text{Prefers high wage/no amenity}
\end{aligned}$$

7.8 Counterfactual: third degree price discrimination

To determine what the firm would pay workers of each type if they could distinguish between types of workers, I use the parameters estimated above and equation ?? to find the profit maximizing wage and profit for each type of worker with and without amenities. Table 12 shows that WTP is inversely related to productivity across types, and that a plurality of the unselected population of potential workers belongs to type 4. It also shows that offering amenities to type 1 and 2 results in lower profit than no amenities since their WTP is below the cost of providing. On the other hand, the firm could increase profits for types 3 and 4 under full information by offering amenities since their WTP is higher than the cost of providing the amenity.

8 Conclusion

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9 Tables and Figures

Table 1: Summary Statistics - Study Sample vs Population

	<i>Survey Sample</i>		<i>Pop, female 18-65 (2016)</i>	
	Mean	N	Mean (all)	Mean (In Labor Force)
Age	32	1,826	35.69	32.51
Married	0.49	1,828	0.65	0.54
Divorced/Widow	0.07	1,828	0.06	0.05
Any kids	0.45	1,828	0.58	0.49
N kids	2.56	827	3.88	3.13
Bachelor's	0.79	1,828	0.19	0.54
Associates	0.95	1,828	0.30	0.72
Amman	0.19	1,828	0.38	0.32
Urban	0.63	1,828	0.87	0.85
Worked in last 3 months	0.48	1,828	0.12	0.61
Worked in last 7 days	0.29	1,828	0.12	0.60

Note: This table reports summary statistics from the survey sample and the Jordan Labor Market Panel Survey from 2016, the most recent publicly available representative labor survey.

Any kids is an indicator for the respondent having any children. N kids reports the number of children conditional on having any children. Bachelors and associates are indicators if the respondent has attained at least that level of education. Amman is an indicator if the respondent lives in Amman, the capital city. Urban is an indicator for living in a self-reported urban area. This includes urban areas in cities smaller than Amman. Worked in the last 3 months/7 days are indicators for having done any amount of wage work or self employment.

Table 2:

	<i>Worked</i>		<i>“Declined”</i>		<i>“Not contacted”</i>		1-2	2-3
	Mean	N	Mean	N	Mean	N		
Age	33	919	32	84	31	823	1.77 * *	0.38
Married	0.45	921	0.48	84	0.53	823	-0.03	-0.06
Any kids	0.42	921	0.42	84	0.49	823	0.00	-0.08
N kids	2.66	387	2.46	35	2.47	405	0.20	-0.01
Bachelor’s	0.72	921	0.87	84	0.86	823	-0.15***	0.01
Associates	0.91	921	1.00	84	0.99	823	-0.09***	0.01
Amman	0.26	921	0.49	84	0.08	823	-0.23***	0.41***
Urban	0.72	921	0.77	84	0.52	823	-0.06	0.26***
Worked in last 3 months	0.62	921	0.38	84	0.35	823	0.24***	0.03
Worked in last 7 days	0.35	921	0.30	84	0.23	823	0.05	0.07
Main current job w/ firm	0.59	534	0.00	25	0.00	227	0.59***	-0.00
Monthly earnings (own, USD PPP)	368	849	291	76	228	740	76.20	63.15
Monthly hh income (USD PPP)	1,631	732	1,622	62	1,381	621	9.05	241.11
Hrs domestic work	20.20	921	22.45	84	23.81	823	-2.25	-1.36
Months worked last 5 years	23.36	921	21.54	84	19.44	823	1.82	2.10
Want a dif job	0.86	534	0.92	25	0.86	227	-0.06	0.06
Career oriented	0.12	913	0.19	83	0.17	818	-0.08 * *	0.02
Financially insecure	0.33	907	0.23	82	0.32	814	0.09*	-0.09*

Note: This table reports summary statistics from three different groups within the survey sample: women who have worked for the firm, women who have likely been contacted by the firm but declined to work, and women who have likely not been contacted. A respondent is marked as having worked if the firm has productivity data for them or if they self-reported to have worked for the firm before. A respondent is determined to likely have declined if they have never worked before and are on the firm’s “inactive” list, which is a list that individuals must select into by stating that they are not interested, or by not replying to 3 separate attempts to contact them. Individuals fall into the not contacted category by having applied, never worked, and not being on the inactive list (meaning they never indicated that they were not interested).

The column marked 1-2 reports the difference between “worked” and “declined” and can be interpreted as the selection effects into employment conditional on a job offer. The column marked 2-3 reports the difference between “declined” and “not contacted” and can be interpreted as the selection effects into being contacted. Consistent with information from the firm about how they contact potential workers, the main differences are geographic.

Any kids is an indicator for the respondent having any children. N kids reports the number of children conditional on having any children. Bachelors and associates are indicators if the respondent has attained at least that level of education. Amman is an indicator if the respondent lives in Amman, the capital city. Urban is an indicator for living in a self-reported urban area. This includes urban areas in cities smaller than Amman. Worked in the last 3 months/7 days are indicators for having done any amount of wage work or self employment.

Main current job with firm is an indicator for the respondent’s main job being with the partner firm. Monthly earnings (own) are the respondents earnings in the past month, scaled from JD to USD PPP by the USD PPP adjustment factor (divide JD by 0.33, see World Bank Data PA.NUS.PRVT.PP). Monthly household income includes all earned and unearned income from any household member in the past month. Hours of domestic work is the number of hours spend on domestic chores, childcare and eldercare in the past 7 days. Months worked last 5 years reports the number of months out of the last 5 years (60 months) that the respondent was employed. Want a dif job is an indicator equal to one if women currently working for a wage selected one of the following options: Two years from now, I’d prefer to 1) be employed in the same job, but with some things changed, or 2) employed in a different job or 3) be self-employed. Career oriented is an indicator equal to one if the respondent reported that building a career was her primary motivation for working besides earning money. (The other options were ‘To gain independence’, ‘To spend my time productively’, ‘None of the above, I only work to earn money’.) Financially insecure is an indicator equal to one if the respondent’s household does not have access to 100 JD / 300 USD PPP in savings and could not easily borrow this amount in an emergency.

Figure 1: WTP for Job Amenities

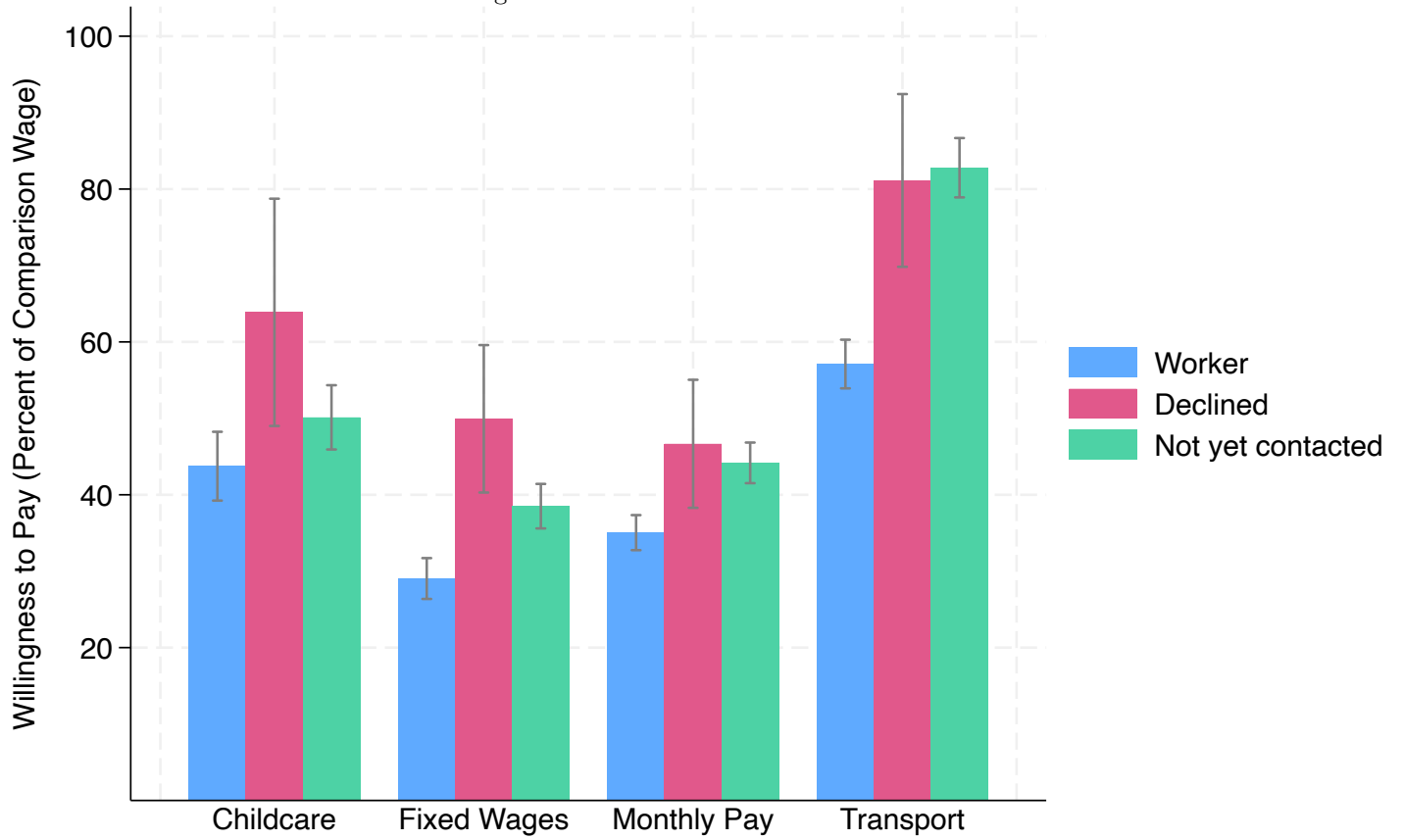


Table 3: Distribution of WTP

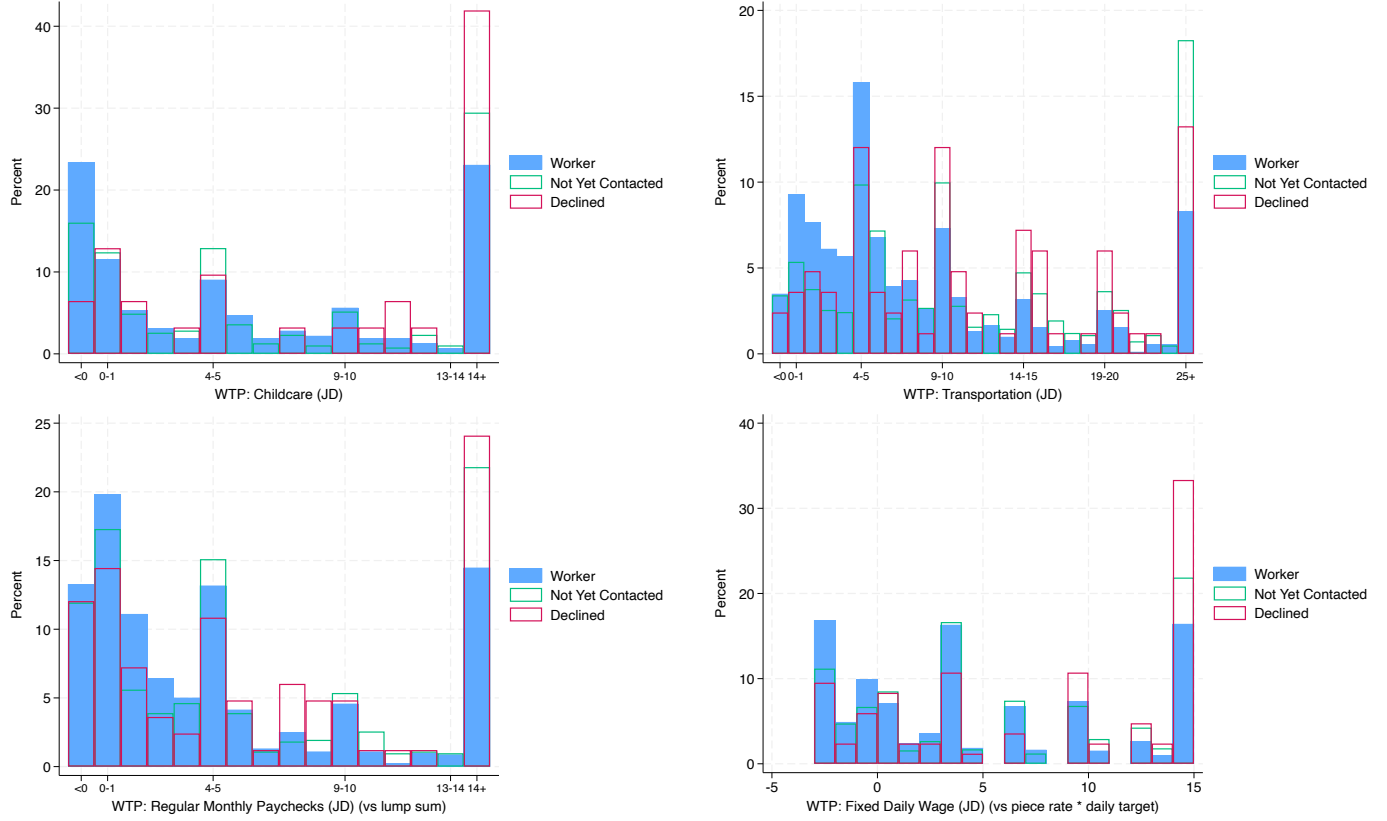


Table 4: Distribution of Productivity Scores

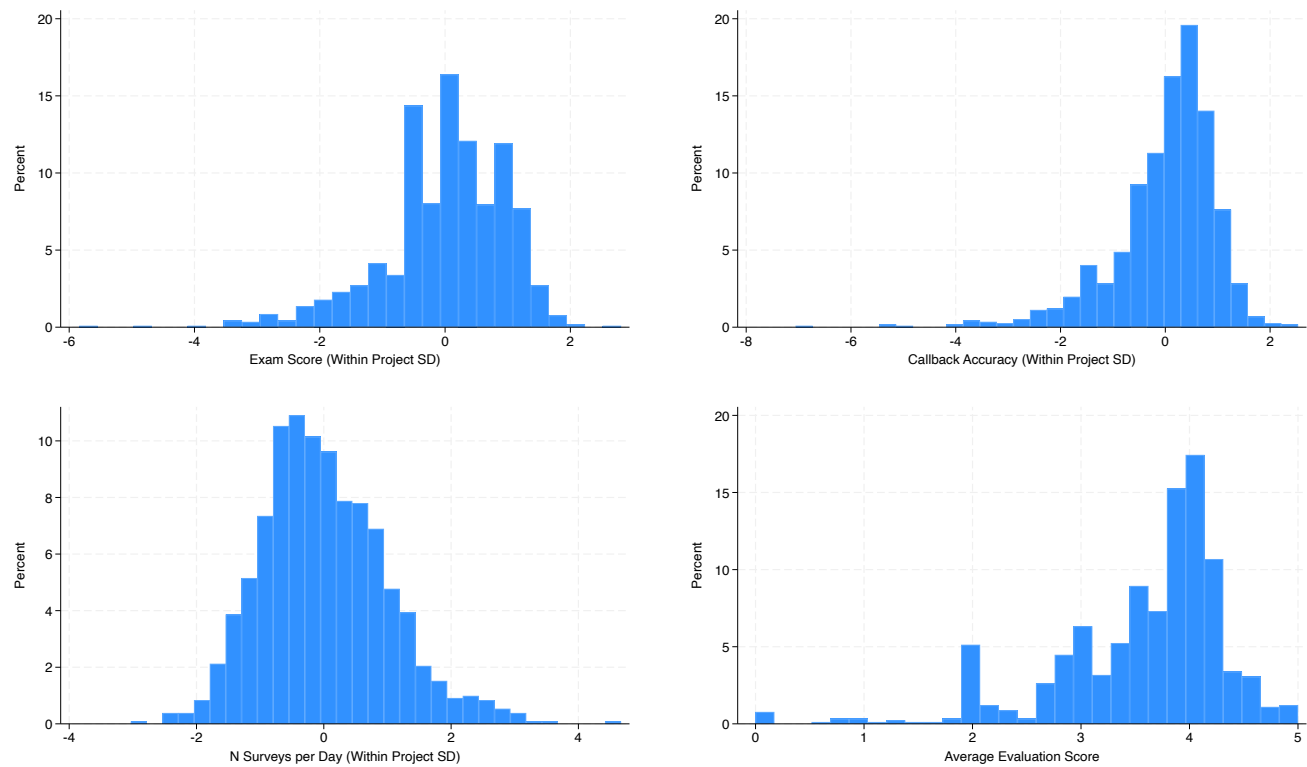


Figure 2: Distribution of WTP and Productivity
 Childcare Transportation

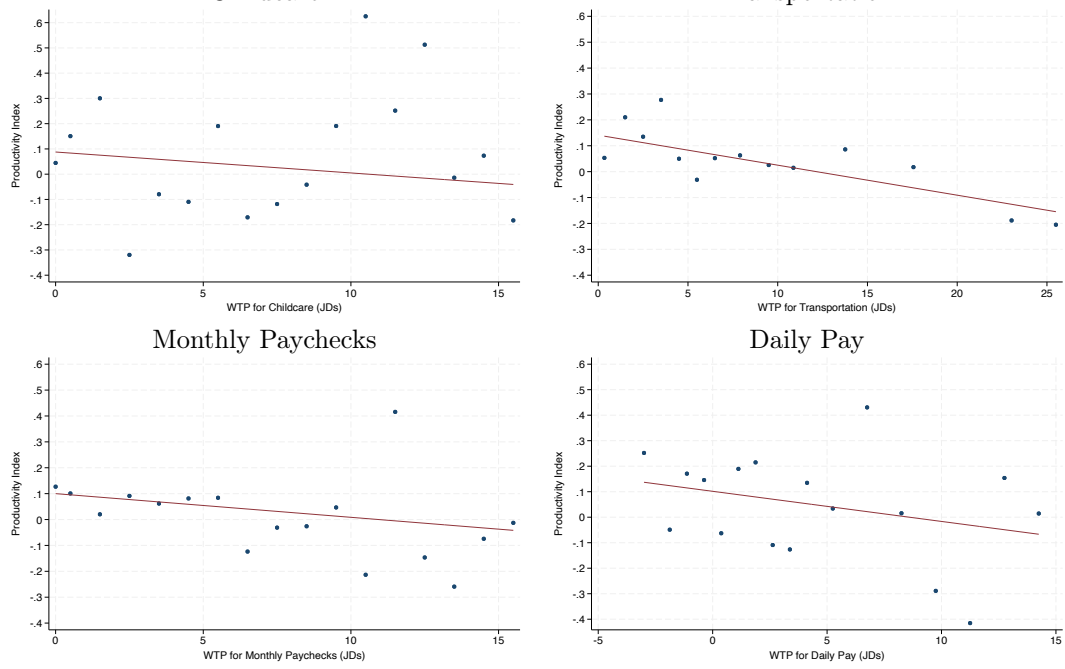


Table 5:

	Transport	Childcare	Monthly Paycheck	Fixed Daily Pay
Urban (=1)	-0.58 (0.44)			
Commute for firm (hours)	0.07 (0.38)			
Commutes to firm job via bus (=1)	-0.39 (0.54)			
Cost of commute to firm job	0.43*** (0.11)			
Feels safe on public transport (SD)	-0.47*** (0.19)			
How often ride is late (SD)	-0.22 (0.26)			
Transport is main barrier (=1)	1.28** (0.66)			
Would use paid childcare (=1)		6.00*** (0.49)		
Mom available (=1)		-2.05*** (0.57)		
Men in fam disparage women who work (SD)		0.71*** (0.22)		
Childcare is biggest barrier (=1)		1.66*** (0.52)		
Man should have final say (SD)		0.39* (0.22)		
wants a different job 2 yrs from now (=1)		1.84** (0.92)		
Motivation for working = career		-0.99 (0.69)		
Minutes to nearest daycare		0.04 (0.03)		
Cost of current paid childcare (JD)		0.01 (0.01)		
<100 savings and can't borrow			0.67** (0.28)	
late paycheck: reduced exp or took formal loan			0.97** (0.42)	
Has formal loan (0/1)			0.76* (0.46)	
Only motivation for working is income				0.58* (0.30)
Worked for firm in last 3 months				-1.83*** (0.42)
Worked in last 3 months	-0.72* (0.39)	-1.16** (0.47)	-0.72*** (0.27)	-0.94*** (0.30)
HH income pc (minus own) (100s USD PPP)	-0.15* (0.08)	0.07 (0.13)	0.03 (0.06)	-0.09 (0.06)
Big 5 neuroticism scale (SD)	0.43 (0.27)	0.82** (0.34)	0.30* (0.19)	0.15 (0.20)

Note:

Table 6: Main Results

	Preferred Productivity Measures			
Transport	-0.00688*** (0.00230)			
Monthly Pay		-0.00600** (0.00274)		
Fixed Wage			-0.00487* (0.00269)	
Childcare				-0.00280 (0.00462)
Constant	0.0531** (0.0231)	0.0309 (0.0221)	0.0206 (0.0192)	0.0106 (0.0350)
Fixed Effects				
Obs	5520	5530	5540	1769

Note: This table reports results from a regression of productivity (SDs) on WTP for the amenity in dinars: $\theta_{ijp} = WTP_i + \epsilon_{ijp}$. Column 1 reports that an individual with WTP 1 dinar higher is expected to be 0.005 SDs less productive. Scaled by the standard deviation of WTP for transportation, this implies that an individual with a 1SD higher WTP for transportation will be on average 0.037 SDs less productive.

For each amenity, the first column reports the regression of productivity on WTP with no controls, and the second column adds fixed effects for locality (smallest administrative district in Jordan) and education level. Controls for these observables reduces the correlation between WTP and productivity by 25 percent for transportation, 30 percent for daily pay, and 10 percent for childcare. Adding fixed effects increases the estimate slightly for the paycheck amenity.

The data is stacked so that the left hand side variable includes all measures of productivity observed for an individual, and the right hand side is WTP. Standard errors are clustered at the individual level.

Table 7: Robustness - Transportation and Monthly Pay

	Preferred Measures		All Measures		Survey Metadata	
Transport	-0.00514** (0.00229)		-0.00515** (0.00219)		-0.00433* (0.00251)	
Monthly Pay		-0.00601** (0.00266)		-0.00433* (0.00260)		-0.00523* (0.00285)
Constant	0.0409* (0.0212)	0.0316 (0.0201)	-0.0154 (0.0200)	-0.0323* (0.0191)	-0.0133 (0.0207)	-0.0193 (0.0192)
Fixed Effects	Location Education	Location Education	Location Education	Location Education	Location Education Nth Project	Location Education Nth Project
Obs	5519	5529	6769	6780	4926	4933

Note:

Table 8: Robustness - Fixed Daily Wages and Childcare

	Preferred Measures		All Measures		Survey Metadata	
Fixed Wage	-0.00315 (0.00261)		-0.00211 (0.00248)		-0.00190 (0.00257)	
Childcare		-0.00258 (0.00483)		0.000470 (0.00450)		0.00382 (0.00494)
Constant	0.0147 (0.0175)	0.0113 (0.0323)	-0.0450*** (0.0164)	-0.0730** (0.0304)	-0.0364** (0.0167)	-0.0769** (0.0321)
Fixed Effects	Location Education	Location Education	Location Education	Location Education	Location Education Nth Project	Location Education Nth Project
Obs	5539	1768	6793	2163	4943	1527

Note:

Table 9: Mediation

	Preferred Productivity Measures			
Transport	-0.00540* (0.00289)	-0.00513* (0.00295)		
Monthly Pay			-0.00608* (0.00333)	-0.00462 (0.00343)
Fixed Effects	Location Education Surveyor	Location Education Surveyor	Location Education Surveyor	Location Education Surveyor
Controls	Demo, income, attitude	Demo, income, attitude + commute time + cost	Demo, income, attitude	Demo, income, attitude + financial precarity
Obs	4250	3568	4394	4315

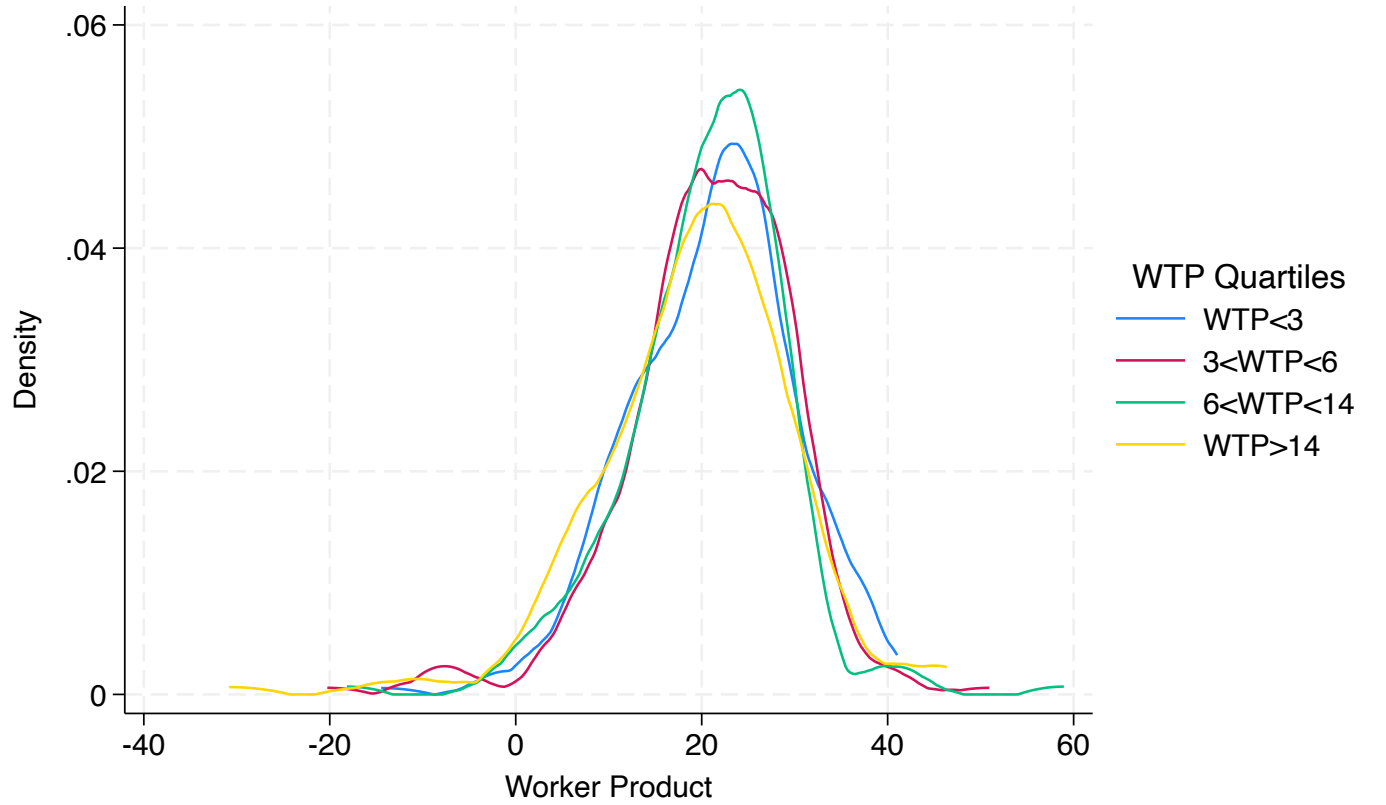
Note:

Table 10: Labor Supply Parameters

Parameter	Estimate
ϕ	5.94 (1.79)
$\underline{b}$	0.84 (0.51)
$\bar{b}$	24.29 (2.17)
$\frac{dF(w)}{dw} \frac{w}{F(w)}$	2.06 (0.22)

Note: This table reports estimates from a nonlinear least squares regression of the log-transformed empirical CDF of reservation wages on the log-transformed mathematical expression of the CDF with the shifted power function function form, of $\ln F(w) = \ln\left(\frac{\ln(w) - \ln(\underline{b})}{\ln(\bar{b}) - \ln(\underline{b})}\right)^\phi$. The underlying reservation wage information comes from a sample of unemployed women in the representative Jordanian Labor Market Panel Survey. The last row reports the labor supply elasticity implied by the parameters estimated above.

Figure 3: Probability Density of Worker Productivity by WTP Quartiles



Note: This figure displays the empirical probability density of rescaled productivity by quartiles of WTP. Productivity is given by the θ_{ijk} index described in section 5, rescaled by the markdown estimated using the firm's profit-maximizing wage first order condition.

Table 11: Firm and Worker Outcomes With and Without Amenities Under Asymmetric Information

	Wage	F(w)	$E[p U_i > 0]$	π	Utility
<i>No amenity</i>					
	13.99	0.39	19.70	2.46	0.04
<i>Amenity</i>					
	9.79	0.50	18.73	2.26	0.06
<i>Difference</i>					
	-4.19***	0.11***	-0.97***	-0.20	0.02

Table 12: Firm and Worker Outcomes With and Without Amenities Under Full Information

Type	v_t	μ_t	ψ_t	(a=0) Wage	(a=1) Wage	(a=0) π_t	(a=1) π_t	(a=0) U_t	(a=1) U_t	Δ Wage	Δ Profit	Δ U
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Note: Share of pop reports what share of the unselected sample (applicants who have not been contacted) belong to each type. WTP_t^{dec} reports the average WTP of individuals of type t. μ_t reports the average productivity (p) of type t. Wage and profit report the profit maximizing wage and profit earned from workers of each type in the full information counterfactual.